ERDŐS PROBLEM #848: AN EXPLICIT SIEVE PROOF AND A KERNEL-CHECKED LEAN FORMALISATION

MACCHACAT

ABSTRACT. Let $A \subseteq \{1, \dots, N\}$ be such that $ab+1$ is not squarefree for all $a, b \in A$ (with $a = b$ allowed). We give a proof that $|A| \le \lfloor (N+18)/25 \rfloor$ for every $N \ge 1$, the number of $n \le N$ with $n \equiv 7 \pmod{25}$; the bound is attained by that residue class. Sawhney proved this for all sufficiently large N without an explicit threshold, and Li gave the first complete proof for all N, with a Lean formalisation. Our argument is independent of Li's: we make every error term in Sawhney's sieve argument explicit, strengthen the lemmas so that the analytic part covers all $N \ge 6 \times 10^5$, and treat the remaining range by a colouring of the graph of compatible integers. The whole theorem is formalised in Lean 4 with Mathlib. Every computation is evaluated by the Lean kernel (`decide +kernel`); no `native_decide` is used, and the only axioms are `propext`, `Classical.choice` and `Quot.sound`. The kernel checks take about five CPU hours and at most 9.3 GB of memory per module.

1. INTRODUCTION

For a positive integer N call a set $A \subseteq \{1, \dots, N\}$ *admissible* if $ab + 1$ is not squarefree for all $a, b \in A$, where $a = b$ is allowed. Since $7 \cdot 7 + 1 = 50$ and $25 \mid ab + 1$ whenever $a \equiv b \equiv 7 \pmod{25}$, the set

$$C_7(N) = \{n \le N : \ n \equiv 7 \pmod{25}\}, \qquad c_7(N) := |C_7(N)| = \left\lfloor \frac{N+18}{25} \right\rfloor,$$

is admissible. Erdős and Sárközy asked whether this is extremal; this is Problem #848 on the Erdős problems site [1].

Theorem 1.1. *For every $N \ge 1$, every admissible $A \subseteq \{1, \dots, N\}$ satisfies $|A| \le c_7(N)$. Hence the maximum size of an admissible set is exactly $c_7(N)$.*

Prior work. Sawhney [2] proved Theorem 1.1 for all sufficiently large N, without an explicit threshold (see also [3, §IV.1]). The first proof for *all* N is due to Li [4, 5], who reduces the extremal problem by a Hall-type equivalence to a completion inequality relative to the progression 7 (mod 25), controls $N \le 5 \times 10^6$ by a prefix-compatible colouring, and formalises the complete theorem in Lean 4. The present work was carried out independently and follows a different route: Sawhney's sieve framework, made fully explicit.

This paper.

(1) *An explicit sieve argument* (Sections 2–5). Every error term in Sawhney's argument is made explicit (Lemmas A′, B, C), with a refined case analysis in which each witness pays the large-prime error only once. Interval certificates in exact rational arithmetic and a tail estimate prove the bound for all $N \ge 1.6 \times 10^8$.
(2) *Strengthened lemmas* (Section 6). Sharper bounds for the large-prime set (a periodic discrepancy estimate), a Pell-type bound counted in a window, Bonferroni truncation, and minimal witnesses lower the analytic threshold to 6×10^5.

Date: October 2026.

(3) *Small N* (Section 7). For small N we exhibit a partition of the vertex set into $c_7(N)$ independent sets of the "compatibility" graph, built incrementally and replayed by an independent checker; this is done up to $N = 4 \times 10^8$ in C++.
(4) *A kernel-checked formalisation* (Section 8). The whole theorem is proved in Lean 4 with Mathlib. All computations are evaluated by the kernel, so neither the Lean compiler nor any external program is trusted. Untrusted programs only produce certificates; the kernel checks them, and the soundness of every checker is proved in Lean.

2. Notation and basic facts

If $a \in A$ then a^2+1 is not squarefree. If $p^2 \mid a^2+1$ then $p \neq 2$ (as $a^2+1 \not\equiv 0 \pmod 4$) and -1 is a square mod p, so $p \equiv 1 \pmod 4$.

Write $A_7 = A \cap (7+25\mathbb{Z})$, $A_{18} = A \cap (18+25\mathbb{Z})$ and $A^* = A \setminus (A_7 \cup A_{18})$. A *U1 prime* is a prime $p \equiv 1 \pmod 4$ with $p \geq 13$. For a U1 prime p let $\pm r_p$ be the two solutions of $x^2 \equiv -1 \pmod{p^2}$ and $R_p = \{x : x \equiv \pm r_p \pmod{p^2}\}$. An integer a is *compatible* with b if $\mu(ab+1) = 0$.

(F1) If $x \in A^*$, then $p^2 \mid x^2+1$ for some U1 prime p, because $25 \mid x^2+1$ iff $x \equiv \pm 7 \pmod{25}$.
(F2) If $a \in A_7 \cup A_{18}$ and $b \in A^*$, then $25 \nmid ab+1$.
(F3) If $a \in A_7$ and $b \in A_{18}$, then $25 \nmid ab+1$, since $7 \cdot 18 + 1 = 127$.
(F4) If a, b are odd, then $4 \mid ab+1$ iff $a \not\equiv b \pmod 4$; if a or b is even, $ab+1$ is odd.
(F5) If $p \nmid b$, then $p^2 \mid ab+1$ iff $a \equiv c_p(b) := -b^{-1} \pmod{p^2}$. For a U1 prime p, $c_p(b) \equiv \pm r_p$ iff $b \equiv \pm r_p$.
(F6) $\rho^*(b) := \max_{(c,b)=1} \#\{s \bmod b : s^2 \equiv c\} \leq 2^{\omega_{\mathrm{odd}}(b)} \cdot \{1,2,4\}$, the last factor according as $v_2(b) \leq 1, = 2, \geq 3$.

3. Sieve lemmas

3.1. **Lemma A′: one witness, shared sets.** Fix a *witness* $1 \leq b \leq N$ and parameters $5 \leq z \leq p_c < Y$. Define

$$\mathrm{Mid}(b) = \{a \leq N : \exists p \in (p_c, Y],\ p \nmid b,\ p^2 \mid ab+1\}, \qquad \mathrm{Ex}(b) = \{a \leq N : \exists p > Y,\ p^2 \mid ab+1\}.$$

Lemma 3.1 (A′). *(i)* $|\mathrm{Mid}(b)| \leq \sum_{p_c < p \leq Y} (N/p^2 + 1)$.
(ii) $|\mathrm{Ex}(b)| \leq \rho^*(b)\big(2(N+1)/Y + (Nb+1)/Y^2\big)$.
(iii) Let every prime factor p of q satisfy $p^2 \mid q$, let T be a set of classes mod q with $p^2 \nmid tb+1$ for all $t \in T$, $p \mid q$; let $\mathcal{P} = \{p \nmid qb\}$ and $k = \#\{p \in \mathcal{P} : p \leq z\}$. With $S = \{a \leq N : a \bmod q \in T\}$,

$$\#\{a \in S :\ a \text{ compatible with } b\} \leq |T|\Big[\frac{N}{q}\Big(1 - \prod_{p \in \mathcal{P},\, p \leq z}\Big(1 - \frac{1}{p^2}\Big)\Big) + 2^k - 1\Big] + \sum_{\substack{z < p \leq p_c \\ p \in \mathcal{P}}} |T|\Big(\frac{N}{qp^2} + 1\Big) + |\mathrm{Mid}(b) \cap S| + |\mathrm{Ex}(b) \cap S|.$$

Proof. If a is compatible with b, pick p with $p^2 \mid ab+1$; then $p \nmid b$, and $p \nmid q$ since otherwise $ab+1 \equiv tb+1 \not\equiv 0 \pmod{p^2}$. Primes $p \leq z$ are handled by inclusion–exclusion in each class t (error $\leq 2^k - 1$); primes $z < p \leq p_c$ by counting one class mod qp^2; primes $p_c < p \leq Y$ give $\mathrm{Mid}(b)$, and each contributes at most N/p^2+1 values of a by (F5). For $p > Y$ write $ab+1 = p^2 m$ with $m \leq M = \lfloor (Nb+1)/Y^2 \rfloor$; a is determined by (p, m), the congruence $p^2 m \equiv 1 \pmod b$ puts p into at most $\rho^*(b)$ classes mod b, and $p \leq \sqrt{(Nb+1)/m}$. Summing over m gives (ii). □

The key point is that when one witness b sieves several disjoint parts of A, the contributions of $\mathrm{Mid}(b)$ and $\mathrm{Ex}(b)$ add up to at most $\Sigma(b) := |\mathrm{Mid}(b)|+|\mathrm{Ex}(b)|$: each witness pays the large-prime error once.

3.2. **Lemma B: sets of type A^*.**

Lemma 3.2 (B)**.** *Let $q \in \{25, 50, 100\}$, T a set of τ classes mod q, $13 \le z_B \le p_B \le Y'$ and $k = \#\{$U1 primes $\le z_B\}$. Then $\#\{n \le N : n \bmod q \in T,\ \exists$ U1 $p,\ p^2 \mid n^2+1\}$ is at most*

$$\tau\Big[\frac{N}{q}\Big(1-\prod_{U1,\,p\le z_B}\Big(1-\frac{2}{p^2}\Big)\Big)+3^k-1\Big]+\sum_{U1,\,z_B<p\le p_B}2\tau\Big(\frac{N}{qp^2}+1\Big)$$
$$+\sum_{U1,\,p_B<p\le Y'}2\Big(\frac{N}{p^2}+1\Big)+\Big\lfloor\frac{N^2+1}{Y'^2}\Big\rfloor\big(\log_3 N+1\big).$$

Proof. Small primes by inclusion–exclusion over the disjoint classes $\pm r_p$; medium primes directly. For $p > Y'$ write $n^2+1 = p^2 m$, so (n,p) solves $x^2 - my^2 = -1$. If $x_1 < x_2$ are two solutions then $x_2 + y_2\sqrt{m} = (x_1 + y_1\sqrt{m})(a + c\sqrt{m})$ with $a^2 - mc^2 = 1$, $a, c \ge 1$, and $x_2 \ge a x_1 \ge 3x_1$ unless $m = 3$, which is impossible ($3 \nmid n^2+1$). Hence each m contributes at most $\log_3 N + 1$ values of n. □

(The original form of the Pell term uses the fundamental unit, with coefficient $1/(2\log(1+\sqrt{2}))$; the elementary bound above is the one formalised.)

3.3. **Lemma C: sieving A^* by an even witness.**

Let $b \in A^*$ be even and T_{25} the 23 classes mod 25 other than 7 and 18. For $x \in A^*$ either $x = b$ or some odd prime $p_2 \nmid b$ has $p_2^2 \mid xb+1$, and by (F1) $x \in R_{p_1}$ for a U1 prime p_1. Writing $C_p = \{x \equiv c_p(b) \pmod{p^2}\}$, $U_1^{\le z_1} = \bigcup_{\mathrm{U1},\,p\le z_1} R_p$ and similarly $U_1^{>z_1}$,

$$(1)\quad |A^*| \le |T_{25}\cap U_1^{>z_1}| + \sum_{\substack{p_2\le z_2\\ p_2 \text{ odd}}}\big|T_{25}\cap U_1^{\le z_1}\cap C_{p_2}\big| + \sum_{z_2<p_2\le p_c}\Big(\frac{N}{p_2^2}+1\Big) + |\mathrm{Mid}(b)\cap A^*| + |\mathrm{Ex}(b)\cap A^*|.$$

Let $\mathcal{U}$ be the U1 primes $\le z_1$, $k_1 = |\mathcal{U}|$ and $P = \prod_{q\in\mathcal{U}}(1-2/q^2)$. Each term of (1) is bounded explicitly: the first by Lemma 3.2 with $q = 25$, $\tau = 23$; the term $p_2 = 5$ by $\frac{N}{25}(1-P) + 3^{k_1}$ (it is one whole class mod 25, inside T_{25}); an odd p_2 that is neither 5 nor a U1 prime by $\frac{23}{25}\frac{N}{p_2^2}(1-P) + 3\cdot 3^{k_1}$; a U1 prime p_2 with $b \in R_{p_2}$ by $\mathrm{whole}(p_2) = \frac{23}{25}\frac{N}{p_2^2} + 3$; and a U1 prime with $b \notin R_{p_2}$ by $\mathrm{non}(p_2) = \frac{23}{25}\frac{N}{p_2^2}(1-\mathrm{pr}_{p_2}) + 3\cdot 3^{k'_{p_2}}$, where $\mathrm{pr}_p = \prod_{q\in\mathcal{U}\setminus\{p\}}(1-2/q^2)$ and $k'_p = |\mathcal{U}\setminus\{p\}|$.

The set $S_b = \{\mathrm{U1}\ p : b \in R_p\}$ is not known, so we bound its worst case. Let $q_1 = 13 < q_2 = 17 < q_3 = 29 < \cdots$ be the U1 primes and $g(p) = \mathrm{whole}(p) - \mathrm{non}(p)$.

Lemma 3.3 (C1, C2)**.** *g is strictly decreasing on U1 primes. If $L = \max\{\ell : q_1\cdots q_\ell \le \sqrt{N^2+1}\}$, then for every even $b \le N$*

$$\sum_{p\in S_b,\ p\le z_2} g(p) \le \sum_{j\le L,\ q_j\le z_2} \max(g(q_j), 0).$$

Proof. The monotonicity follows from the explicit form of g on either side of z_1. For the second claim, the primes of S_b are distinct and $p^2 \mid b^2+1$ for each, so $\prod_{p\in S_b} p \le \sqrt{N^2+1}$; the j-th smallest element of S_b is $\ge q_j$, so $|S_b| \le L$ and $g(s_j) \le g(q_j)$. □

4. Case analysis

Write $R(T, q; \text{excl})$ for the terms of Lemma 3.1(iii) other than Mid and Ex, in the worst case $\mathcal{P}$ = all primes outside excl, and $B(\tau, q)$ for the bound of Lemma 3.2. "Lemma C" denotes (1) without its last two terms; together with the shared part of $A_7 \cup A_{18}$ for the same witness, those terms are bounded by $\Sigma(b)$.

Case	Condition	Upper bound
I.a	$A^* = \emptyset$, A_7 or A_{18} empty	$c_7(N)$
I.b	$A^* = \emptyset$, $b \in A_7$, $b' \in A_{18}$	$R(\{18\}, 25; \{5\}) + \Sigma(b) + R(\{7\}, 25; \{5\}) + \Sigma(b')$
II	A^* has an even element b	$R(\{7, 18\}, 25; \{2, 5\}) + \Sigma(b) +$ Lemma C
III.1	A^* odd, with $b_1 \equiv 1$, $b_3 \equiv 3 \pmod 4$	$R(6 \text{ cl.}, 100) + \Sigma(b_1) + R(2 \text{ cl.}, 100) + \Sigma(b_3) + B(23, 50)$
III.2.a	A^* in one class mod 4, $A_7 \cup A_{18}$ odd	$2(N/100 + 1) + R(2 \text{ cl.}, 100) + \Sigma(b) + B(23, 100)$
III.2.b	as III.2.a, but $A_7 \cup A_{18}$ has an even b'	$(N/100 + 1) + R(3 \text{ cl.}, 100) + \Sigma(b) + R(4 \text{ cl.}, 100) + \Sigma(b') + B(23, 100)$

The main coefficients (times 25) are 0.741, 0.784, 0.632, 0.739, 0.686 for I.b, II, III.1, III.2.a, III.2.b; all are < 1. The cases are exhaustive, and the residue conditions used are consequences of (F2)–(F4). For instance in III.1, b_1 sieves the classes $57, 93, 32, 82, 18, 68 \pmod{100}$ of $A_7 \cup A_{18}$ and b_3 the classes $7, 43$; in each of them $4 \nmid tb + 1$ and $25 \nmid tb + 1$, so $\{2, 5\}$ may be excluded.

5. Large N: interval certificates

Lemma 5.1 (D)**.** *Fix parameters on an interval $[N_{lo}, N_{hi}]$. Suppose that for every case and every integer N in the interval the bound has the form $CN + E(N)$ with E nondecreasing. If $C < 1/25$ and $CN_{lo} + E(N_{hi}) < (N_{lo} - 6)/25$, then $|A| < (N - 6)/25 \le c_7(N)$ for every integer $N \in [N_{lo}, N_{hi}]$.*

Proof. $(N - 6)/25 - CN$ is increasing, and $c_7(N) \ge (N + 18 - 24)/25$. □

The parameters (z, p_c, Y, Y'_q, $p_{B,q}$, z_B, z_1, z_2, L) are chosen by a floating-point optimiser; since every choice gives a valid bound, they need no verification. The bounds are then recomputed in exact rational arithmetic: $\sum 1/p^2$ by integer sums of $\lceil K/p^2 \rceil$ up to 2×10^7 and $\sum_{n>P} 1/n^2 \le 1/P$ beyond, $\pi(x)$ exactly up to 2×10^7 and by an explicit upper bound above (in the formalised version, Chebyshev's bound $\theta(x) \le x \log 4$, which is in Mathlib), and logarithms by rational bounds in the safe direction. In the formalised version, 2913 intervals with integer endpoints cover $[1.6 \times 10^8, 10^{60}]$, with worst ratio 0.9891 (the ratio of the bound to $(N_{lo} - 6)/25$).

For $N \ge 10^{60}$ the parameters are fixed functions of N (for instance $Y = N^{0.7}$ and $p_c = p_B = 2000$), $\rho^*(b) \le 2N^{0.195}$ by Robin's bound on $\omega(n)$, and C is constant. Each term of $E(N)/N$ has the form $aN^{\alpha-1}(\log N)^\gamma$ with $\alpha \le 0.8$ and is decreasing for $N \ge 10^{60}$; at $N = 10^{60}$ the ratios are at most 0.826 in every case. The formalised tail uses $t = \lfloor N^{1/4} \rfloor$, $Y = \lfloor N/t \rfloor$ and the bound $\rho(b)^5 \cdot (3 \cdot 5 \cdots 31) \le 1024 \cdot 32^{10}\, b$, valid for all b, instead of Robin's inequality. This proves Theorem 1.1 for $N \ge 1.6 \times 10^8$.

6. Strengthened lemmas: the analytic part from 6×10^5

6.1. **The large-prime set.** Fix a witness b and $Y > p_c$; put $M = \lfloor (Nb + 1)/Y^2 \rfloor$, $X_m = \sqrt{(Nb + 1)/m}$ and $r_b(m) = \#\{s \bmod b : s^2 m \equiv 1 \pmod b\}$.

Lemma 6.1 (E). *(E0)* $|\mathrm{Ex}(b)| \le \sum_{m\le M} r_b(m)\big(\lfloor (X_m - Y)/b\rfloor + 1\big)$.
(E1) $\sum_{m\le M}\big((X_m - Y)/b + 1\big) \le (N+1)/Y + M$; *hence* $|\mathrm{Ex}(b)| \le \rho^*(b)\big((N+1)/Y + M\big)$.
(E2) Let $e = \min(v_2(b), 3)$, Q *a set of odd primes dividing* b, $P = 2^e\prod Q$, $g_P = r_P$, $\alpha_P = \varphi(P)/P$ *and* $\beta_P = \max_{0\le R<P}\big(\sum_{m\le R} g_P(m) - \alpha_P R\big)$. *Then*

$$|\mathrm{Ex}(b)| \le 2^{\omega_{\mathrm{odd}}(b)-|Q|}\Big[\alpha_P\Big(\frac{N+1}{Y} + M\Big) + \beta_P\Big(\frac{\sqrt{Nb+1}}{b} + 1\Big)\Big].$$

Sketch. (E0) $a \mapsto (p, m)$ with $ab + 1 = p^2 m$ is injective and p lies in $r_b(m)$ classes mod b in $(Y, X_m]$. (E1) follows from $\sum_{m\le M} m^{-1/2} \le 2\sqrt{M} - 1$ and $(K - \sqrt{M})^2 \ge 0$ with $K = \sqrt{Nb+1}/Y$. For (E2), $r_b \le 2^{\omega_{\mathrm{odd}}(b)-|Q|} g_P$ by the Chinese remainder theorem; $\sum_{m \bmod P} g_P(m) = \varphi(P)$, so $G(R) = \sum_{m\le R} g_P(m) \le \alpha_P R + \beta_P$; Abel summation against the nonincreasing $f(m) = \lfloor (X_m - Y)/b\rfloor + 1$ and (E1) finish the proof. □

To use (E2) uniformly over unknown witnesses, a witness $b \le x$ is described by its *structure* (e, S, u): $e = \min(v_2(b), 3)$, S the small odd primes dividing b, u the number of larger odd prime factors. Each structure has a least possible value $b_{\min}$, and the bound is maximised over the structures with $b_{\min} \le x$, using the table of $(P, \varphi(P), P\beta_P)$.

Lemma 6.2 (M). *A class mod* p^2 *with* $p > \sqrt{N}$ *has at most one element in* $[1, N]$, *so* $|\mathrm{Mid}(b)| \le \sum_{p_c<p\le \min(Y,\sqrt{N})}(N/p^2 + 1) + \#\{p : \max(p_c, \sqrt{N}) < p \le Y\}$.

6.2. **The U1 tail.**

Lemma 6.3 (H). *For a U1 prime* p, $\#\{n \le N : p^2 \mid n^2 + 1\} \le 2(N/p^2 + 1)$; *it is* ≤ 2 *if* $p^2 > N$ *and* ≤ 1 *if* $p^2 > 2N$.

Lemma 6.4 (P). *Let* $Y_q \ge 5$, $M = \lfloor (N^2+1)/Y_q^2\rfloor$, *and call* m admissible *if* $m \bmod 8 \in \{1, 2, 5\}$ *and* $3, 7, 11, 19, 23 \nmid m$. *Then*

$$\#\{n \le N : \exists\, p \ge Y_q,\ p^2 \mid n^2 + 1\} \le \sum_{\substack{m\le M\\ m\ admissible}} \Big(1 + \Big\lfloor \frac{\log(N/x_m)}{\log g_m}\Big\rfloor_+\Big),$$

where $x_m = \sqrt{mY_q^2 - 1}$ *and* $g_m = \max(3, 2\sqrt{m})$.

Proof. With $n^2 + 1 = p^2 m$ we have $m \equiv n^2 + 1 \in \{1, 2, 5\} \pmod 8$, no prime $\equiv 3 \pmod 4$ divides m, and $n \ge x_m$. Consecutive solutions of $x^2 - my^2 = -1$ grow by a factor $\ge g_m$, as in Lemma 3.2. □

6.3. **Bonferroni truncation and minimal witnesses.**

Lemma 6.5 (Bf). *Let* D *be a set whose intersection with each class mod* qQ (Q *odd, coprime to* q) *has size in* $[L/(qQ) - \mu, L/(qQ) + \mu]$, *let* $p_1, \dots, p_k$ *be primes coprime to* q, *and let* E_i *consist of* c *classes mod* p_i^2, $v_i = c/p_i^2$. *For odd* d, *the number of elements of* D *in one class mod* q *lying in* $\bigcup E_i$ *is at most* $\frac{L}{q}\sum_{j=1}^{d}(-1)^{j+1} e_j(v_1, \dots, v_k) + \mu\sum_{j=1}^{d}\binom{k}{j}c^j$, *and the main term is nondecreasing in the set of primes.*

Minimal witnesses. Each witness is taken to be the least element of its part, so the main terms may use $L = N - x + 1$ in place of N. The witnesses are grouped into buckets $[x_a, x_b]$ with $x_b = \max(x_a + 1, \lfloor 1.3x_a\rfloor)$, and in each bucket $\Sigma(b)$ is replaced by a bound $\mathrm{Sh}(x_b)$ valid for all $b \le x_b$. When two witnesses sieve each other's parts, the terms can be re-paired so that each pair is bounded by twice the maximum over buckets.

6.4. **Revised Lemma C.** Lemma C only uses that A^* lies in a domain D whose classes mod odd Q have $L_{\text{eff}}/Q \pm \mu$ elements. In case II, with b the least even element of A^*, $L_{\text{eff}} = N-(b-1)/2$ and $\mu = 2$; for an odd witness b and $x \equiv b \pmod 4$ we have $xb+1 \equiv 2 \pmod 4$, which gives Lemma C′ on $(b+4\mathbb{Z}) \cap [b, N]$ with $L_{\text{eff}} = (N-b+1)/4$. For each odd $p_2 \neq 5$, $\text{non}(p_2)$ is the least of three valid bounds: truncated inclusion–exclusion without the restriction mod 25, full inclusion–exclusion as "all classes minus two", and truncated inclusion–exclusion class by class. (Truncated bounds are upper bounds only, so they cannot be used for a subtracted part.) Lemma C1 is replaced by:

Lemma 6.6 (C2′)**.** *If $b \le x$, $n_S = \max\{\ell : q_1 \cdots q_\ell \le \sqrt{x^2+1},\ q_\ell \le z_2\}$ and $g(p) = \max(\text{whole}(p) - \text{non}(p), 0)$, then $\sum_{p \in S_b,\, p \le z_2}(\text{whole}(p) - \text{non}(p))$ is at most the sum of the n_S largest values of g. Equivalently, for every $\tau \ge 0$ it is at most $n_S\tau + \sum_p \max(g(p) - \tau, 0)$.*

6.5. **Case analysis and certificate.** The case analysis of Section 4 is refined (III.2 is split according to which of the classes $\{7, 32, 82\}$, $\{18, 68, 43\}$ are present, with two or three witnesses), and every bound is nondecreasing in N for fixed parameters, so it suffices to check $\text{bound}(N_{hi}) < (N_{lo}-6)/25$ on each interval. An exact certificate (C with GMP) passes on 396 intervals covering $[6\times10^5, 1.6\times10^8]$, with worst ratio 0.9594; a second implementation in Python rationals agrees.

7. Small N: a checked colouring

Let $V = \{n : \mu(n^2+1) = 0\}$ and let G be the graph on V with an edge uv when $\mu(uv+1) = 0$. Admissible sets are the cliques of $G[V \cap [1, N]]$, so it suffices to partition $V \cap [1, N]$ into $c_7(N)$ independent sets.

We build such partitions incrementally in n: a new class is opened only when $n \equiv 7 \pmod{25}$, and every other vertex is placed into an existing class, if necessary after moving one conflicting element to another class. Every operation is logged, and an independent program replays the log, recomputing V by a different sieve and squarefreeness by complete factorisation. Up to $N = 4\times10^8$ this gives $|V| = 42{,}063{,}559$ vertices in $16{,}000{,}000 = c_7(4\times10^8)$ classes; both replays pass.

8. The Lean formalisation

8.1. **Statement.** The formal statement is

```
def Admissible (A : Finset ℕ) : Prop :=
  ∀ a ∈ A, ∀ b ∈ A, ¬ Squarefree (a * b + 1)
def c7 (N : ℕ) : ℕ := (N + 18) / 25
def Statement (N : ℕ) : Prop :=
  ∀ A : Finset ℕ, A ⊆ Finset.Icc 1 N → Admissible A → A.card ≤ c7 N
theorem Erdos848.erdos848 : ∀ N, Statement N
```

together with `erdos848_max'` (the maximum is exactly $c_7(N)$) and the form used by the Formal Conjectures project. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

8.2. **Trust.** Every computation is a Boolean checker evaluated by `decide +kernel`. No `native_decide` is used, so neither the compiler nor the interpreter is trusted. The data (interval parameters, bucket bounds, witness structures, colouring operations) are produced by untrusted programs; the kernel checks the data, and for each checker a soundness theorem connects its output to the real bounds of the lemmas above. As an independent check, `leanchecker`, which is shipped with Lean, re-checks the declarations of each module of the project in the kernel, module by module; every module that contains declarations passes. The split in Lean is:

- $N \le 986{,}746$: the colouring, replayed by the kernel;
- $986{,}747 \le N \le 1.6 \times 10^8$: Section 6, 346 intervals, worst ratio 0.9306;
- $N \ge 1.6 \times 10^8$: Section 5, 2913 certificate rows and the tail.

8.3. **Kernel arithmetic.**

- *Primes.* For $n < 4473^2$, n is prime iff $\gcd(n, \prod_{p \le 4472} p) = 1$. The prime counts and the sums $\sum \lceil 10^{30}/p^2 \rceil$ at the 10,998 points needed below 2×10^7 are verified by binary-split segment sums. Lookups use a search tree written as a literal and checked against the table by one in-order traversal.
- *Squarefreeness* for $n < 10007^3$: with $g = \gcd(n, \prod_{p \le 10^4} p)$, n is squarefree if $\gcd(g, n/g) = 1$ and n/g is not a square > 1.
- *Lemma C in fixed point.* The sums over the primes $p \le z_2$ are computed with scale 2^{40}; per-prime constants are rounded once, up, or down where they are subtracted, and each bucket then needs only natural-number arithmetic. The threshold τ of Lemma C2′ is the n_S-th largest rounded gain; any $\tau \ge 0$ is sound. Exact rational sums would involve denominators that are products of many prime squares; avoiding them made the evaluation of Lemma C about 50 times faster and cut the memory of an interval check by an order of magnitude.
- *The table of Lemma E2* is restricted to the 268 entries with period $P \le 3000$, and each β_P is verified over one period; witness structures distinguish odd prime factors ≤ 23.
- *The colouring* for $N \le 986{,}746$: an untrusted first-fit proposer with augmenting moves emits 117,389 operations (open a class, place a vertex, move a vertex). The kernel replays them on classes stored in a binary tree of depth 16, in 40 windows whose end states are compared with tree-literal checkpoints. The vertex set is certified by roots of $x^2 \equiv -1 \pmod{p^2}$ for every prime $p \equiv 1 \pmod 4$, $p \le 986{,}746$, together with a proof that these are all such primes.

8.4. **Cost.** The kernel keeps every reduction of a declaration until the declaration is checked, so large checks are split into many declarations and reassembled by small lemmas. A full build takes about five CPU hours (about one hour on a 20-core machine). The peak memory per module is at most 3.5 GB for an interval of Section 6, at most 9.3 GB for a colouring window, and at most 7.6 GB otherwise.

9. Reproducibility

All code, certificates and the Lean project are in the accompanying repository, `https://github.com/MacchaCat/erdos848`. The certificates of Sections 5 and 6 can be re-checked with the exact Python and C programs; the colouring logs up to 4×10^8 are published with SHA-256 checksums; and `lake build` in the Lean project re-checks everything in the kernel.

Disclosure and review status

The proof, the code and the Lean formalisation were produced mainly by an AI system (Claude, Anthropic), under the direction of the author, who set the goals and chose the directions, with a small number of manual interventions. The written argument went through several rounds of AI review and an independent re-implementation of all bounds from the text alone; the issues found were fixed. The complete argument is checked by the Lean kernel. The mathematics has not yet been reviewed by human experts, and readers should also check that the formal statement above is the intended one.

This manuscript is released under the Creative Commons Attribution 4.0 International licence (CC BY 4.0); the code and the Lean project are released under the Apache License 2.0.

References

[1] T. F. Bloom, *Erdős Problem #848*, https://www.erdosproblems.com/848.

[2] M. Sawhney, note on Erdős Problem #848, https://www.math.columbia.edu/~msawhney/Problem_848.pdf.

[3] arXiv:2511.16072, Section IV.1.

[4] A. C. Li, *The Exact Extremal Bound in Erdős Problem 848*, August 2026, https://ssrn.com/abstract=7230480.

[5] A. C. Li, *erdos-848-squarefree-product*, https://github.com/crabsatellite/erdos-848-squarefree-product.

[6] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, LNCS 12699 (2021), 625–635.

[7] The mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381.